Analysis

PENG

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1 Introduction

Definition 1.1 (analysis)**.** *Analysis of real numbers, sequences and series of real numbers, real-valued functions, and geometry [1, 2].*

2 Natural Number

Definition 2.1 (natural number)**.** *A natural number is any element of the set*

$$\mathbb{N} := 0, 1, 2, 3, \ldots \tag{2.1}$$

Axiom 2.1 (0)**.** *0 is a natural number.*

Axiom 2.2 (successor)**.** *If n is a natural number, then $n{+}{+}$ is also a natural number.*

Definition 2.2 (notation)**.** $1 := 0{+}{+}$, $2 := 1{+}{+}$, $3 := 2{+}{+}$, ...

Axiom 2.3 ($n{+}{+} \neq 0$)**.** *0 is not the successor of any natural number n.*

Axiom 2.4 ($n{+}{+} \neq m{+}{+}$)**.** *Different natural numbers must have different successors.*
If $n \neq m$, then $n{+}{+} \neq m{+}{+}$. Equivalently, if $n{+}{+} = m{+}{+}$, then $n = m$.

Axiom 2.5 (principle of mathematical induction)**.** *Let $P(n)$ be any property pertaining to a natural number n. Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n{+}{+})$ is also true. Then $P(n)$ is true for every natural number n.*

Lemma 2.1 (recursive definition of sequence)**.** *$a_0 = c$ and $a_{n{+}{+}=f_n(a_n)}$ for each natural number n.*

2.1 Addition

Definition 2.3 (addition)**.** *Define $0 + m := m$. Suppose inductively that we have defined $n + m$. Then $(n{+}{+}) + m := (n + m){+}{+}$.*

$a_n = n + m$ and $f_n(a_n) = a_n{+}{+}$

Lemma 2.2. $n + 0 = n$

Lemma 2.3. $n + (m{+}{+}) = (n + m){+}{+}$

Proposition 2.1 (addition is commutative)**.**

$$n + m = m + n \tag{2.2}$$

Proposition 2.2 (addition is associative)**.**

$$(a + b) + c = a + (b + c) \tag{2.3}$$

Proposition 2.3 (cancellation law)**.** *If $a + b = a + c$, then $b = c$.*

Definition 2.4. *A natural number n is positive if and only if $n \neq 0$.*

Lemma 2.4. *If a is a positive natural number and b is a natural number, then $a + b$ is positive.*

Lemma 2.5. *If $a + b = 0$, then $a = 0$ and $b = 0$.*

Lemma 2.6. *If a is a positive number, then there exists exactly one natural number b such that $b{+}{+} = a$.*

Definition 2.5 (order)**.**

- *$n \geq m$ or $m \leq n$ if and only if $n = m + a$ for some natural number a*
- *$n > m$ or $m < n$ if and only if $n \geq m$ and $n \neq m$*

Proposition 2.4 (properties of order)**.**

- *reflexive: $a \geq a$*
- *transitive: If $a \geq b$ and $b \geq c$, then $a \geq c$*
- *antisymmetric: If $a \geq b$ and $b \geq a$, then $a = b$*
- *addition preserve order: $a \geq b$ if and only if $a + c \geq b + c$*
- *$a < b$ if and only if $a{+}{+} \leq b$*

- $a < b$ *if and only if* $b = a + d$ *for some positive number* d

Proposition 2.5 (trichotomy of order)**.** *Exactly one of following statement is true:* $a < b$, $a = b$, *or* $a > b$.

Proposition 2.6 (strong principle of induction)**.** *Let* m_0 *be a natural number. Suppose that for each* $m \geq m_0$, *if* $P(m')$ *is true for all* $m_0 \leq m' < m$, *then* $P(m)$ *is also true. Then* $P(m)$ *is true for all* $m \geq m_0$.

Example 2.1 (Fibonacci sequence)**.**

$$F_0 = 0 \tag{2.4}$$
$$F_1 = 1 \tag{2.5}$$
$$F_n = F_{n-1} + F_{n-2} \tag{2.6}$$

2.2 Multiplication

Definition 2.6 (multiplication)**.** *Define* $0 \times m := 0$. *Suppose inductively that we have defined* $n \times m$. *Then* $(n{+}{+}) \times m := (n \times m) + m$.

Lemma 2.7. $n \times 0 = 0 \times n$

Lemma 2.8. $n \times (m{+}{+}) = (n \times m) + n$

Proposition 2.7 (multiplication is commutative)**.**

$$n \times m = m \times n \tag{2.7}$$

Lemma 2.9 (zero divisor)**.** $n \times m = 0$ *if and only if at least one of* n *and* m *is equal to 0. If* n *and* m *are both positive, then* nm *is also positive.*

Proposition 2.8 (distributive law)**.**

$$a(b + c) = (b + c)a = bc + ca \tag{2.8}$$

Proposition 2.9 (multiplication is associative)**.**

$$(ab)c = a(bc) \tag{2.9}$$

Lemma 2.10 (multiplication preserve order)**.** *If* $a < b$ *and* c *is positive, then* $ac < nc$.

Proposition 2.10 (cancellation law)**.** *If* $ac = bc$ *and* c *is positive, then* $a = b$.

Proposition 2.11 (Euclid's division lemma)**.** *Let* n *be a natural number and* q *be a positive natural number. Then there exist natural numbers* m *and* r, *such that* $0 \leq r < q$ *and* $n = mq + r$.

Proof. $0 = 0 \times q + 0$ and $0 \leq r < q$. Suppose that $n = mq + r$ and $0 \leq r < q$. Then $n{+}{+} = mq + (r{+}{+})$ and $0 \leq r{+}{+} \leq q$. If $r{+}{+} = q$, $n{+}{+} = mq + q = (m + 1)q + 0$. If $r{+}{+} < q$, $n{+}{+} = mq + r{+}{+}$. □

Definition 2.7 (Exponentiation)**.** *Define* $m^0 := 1$, *in particular* $0^0 := 1$. *Suppose recursively that* m^n *has been defined for some natural number* n, *then define* $m^{n++} := m^n \times m$.

3 Set Theory

Definition 3.1 (set)**.** *Define a set* A *to be any unordered collection of objects. If* x *is an element of* A, *then* $x \in A$, *otherwise* $x \notin A$.

Axiom 3.1 (sets are objects)**.** *If* A *is a set, then* A *is also an object.*

Axiom 3.2 (equality of sets)**.** $A = B$ *if and only if every element of* A *is an element of* B *and vice versa.*

Axiom 3.3 (empty set)**.** *There exists an empty set* $\emptyset$ *which contains no elements.*

Lemma 3.1 (single choice)**.** *If* A *is a non-empty set, then there exists an object* x *such that* $x \in A$.

Axiom 3.4 (singleton and pair sets)**.** *If a is an object, then there exists a singleton set $\{a\}$ whose only element is a. If a and b are objects, then there exists a pair set $\{a, b\}$ whose only elements are a and b.*

Axiom 3.5 (pairwise union)**.** *If A and B are two sets, then there exists a union of A and B set $A \cup B$, which consists of all the elements which belong to A or B or both.*

$$x \in A \cup B \iff x \in A \text{ or } x \in B \tag{3.1}$$

Proposition 3.1 (properties of union)**.**

- $\{a, b\} = \{a\} \cup \{b\}$
- *commutative:* $A \cup B = B \cup A$
- *associative:* $(A \cup B) \cup C = A \cup (B \cup C)$

Definition 3.2 (subsets)**.** *A is a subset of B, denoted $A \subseteq B$, if and only if every element of A is also an element of B.*

A is a proper subset of B, denoted $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Axiom 3.6 (axiom of specification)**.** *Let A be a set and $P(x)$ be a property pertaining to x for each $x \in A$. Then there exists a set $\{x \in A : P(x) \text{ is true}\}$ whose elements are precisely the elements x in A for which $P(x)$ is true.*

Definition 3.3 (intersections)**.** *The intersection of two sets is defined to be the set*

$$A \cap B := \{x \in A : x \in B\} \tag{3.2}$$

Definition 3.4 (difference sets)**.** *Define $A - B$ or $A \backslash B$ to be set A with elements of B removed*

$$A \backslash B := \{x \in A : x \notin B\} \tag{3.3}$$

Axiom 3.7 (replacement)**.** *If $P(x, y)$ is a property pertaining to x and y, such that for each $x \in A$ there is at most one y for which $P(x, y)$ is true. Then there exists a set $\{y : P(x, y) \text{ is true for some } x \in A\}$.*

Axiom 3.8 (infinity)**.** *There exists a set $\mathbb{N}$, whose elements are called natural numbers, as well as an object 0 in $\mathbb{N}$, and an object $n{+}{+}$ assigned to every $n \in \mathbb{N}$, such that the Peano axioms hold.*

3.1 Cartesian Product

Definition 3.5 (ordered pair)**.** *If x and y are any objects, define the ordered pair (x, y) to be a new object. Two ordered pairs (x, y) and (x', y') are equal if and only if both their components match*

$$(x, y) = (x', y') \iff x = x' \text{ and } y = y'. \tag{3.4}$$

Definition 3.6 (Cartesian product)**.** *If X and Y are sets, then define Cartesian product $X \times Y$ to be the collection of ordered pair*

$$X \times Y = \{(x, y) : x \in X, y \in Y\} \tag{3.5}$$

Definition 3.7 (ordered sequence of n elements)**.** *If $(x_1, \ldots, x_n)$ is an ordered n-tuple, define Cartesian product $X_1 \times \cdots \times X_n$ by*

$$\prod_{1 \leq i \leq n} X_i := \{(x_i)_{1 \leq i \leq n} : x_i \in X_i \text{ for all } 1 \leq i \leq n\} \tag{3.6}$$

Lemma 3.2 (finite choice)**.** *Let X_i be a non-empty set, then there exists an n-tuple $(x_i)_{1 \leq i \leq n}$ such that $x_i \in X_i$ for all $1 \leq i \leq n$.*

4 Function

Definition 4.1 (function)**.** *Let X, Y be sets, and let $P(x, y)$ be a property pertaining to an object $x \in X$ and an object $y \in Y$, such that for every $x \in X$, there is exactly only one $y \in Y$ for which $P(x, y)$ is true. Then the function $f : X \to Y$ is defined by P on the domain X and codomain (range) to be the unique object $f(x) \in Y$ for which $P(x, f(x))$ is true.*

Definition 4.2 (equality of functions)**.** *Two functions $f : X \to Y$ and $g : X' \to Y'$ are equal if their domains and codomains agree, and $f(x) = g(x)$ for all $x \in X$.*

Definition 4.3 (composition)**.** *Let $f : X \to Y$ and $g : Y \to Z$ be two functions, such that the codomain of f is the same set as the domain of g. Then define composition $g \circ f : X \to Z$ to the function*

$$(g \circ f)(x) := g(f(x)) \tag{4.1}$$

Definition 4.4 (injective)**.** *A function is one-to-one if every element in X maps to different element in Y*

$$x \neq x' \Longrightarrow f(x) \neq f(x') \tag{4.2}$$

Equivalently,

$$f(x) = f(x') \Longrightarrow x = x' \tag{4.3}$$

Definition 4.5 (surjective)**.** *A function is onto if every element in Y comes from applying f to some element in X.*

$$\text{For every } y \in Y, \text{ there exists } x \in X \text{ such that } f(x) = y. \tag{4.4}$$

Definition 4.6 (bijective)**.** *A function is bijective or invertible if it is both one-to-one and onto.*

4.1 Images

Definition 4.7 (images of sets)**.** *If $f : X \to Y$ is a function from X to Y, and S is a subset of X, define $f(S)$ to be the set*

$$f(S) := \{f(x) : x \in S\} \tag{4.5}$$

Definition 4.8 (inverse images)**.** *If U is a subset of Y, define the set $f^{-1}(U)$ to be the set*

$$f^{-1}(U) := \{x \in X : f(x) \in U\} \tag{4.6}$$

Axiom 4.1 (power set axiom)**.** *Let X and Y be sets. Then there exists a set Y^X which consists of all the functions from X to Y*

$$f \in Y^X \Longleftrightarrow f \text{ is a function with domain } X \text{ and codomain } Y. \tag{4.7}$$

Axiom 4.2 (union)**.** *Let A be a set whose elements are sets. Then there exists a set $\bigcup A$*

$$x \in \bigcup A \Longleftrightarrow x \in S \text{ for some } S \in A. \tag{4.8}$$

5 Cardinality of Sets

Definition 5.1 (ordinal and cardinal)**.** *The cardinals are One, Two, ..., and are used to count how many things there are in a set.*

THe ordinals are First, Second, ..., and are used to order a sequence of objects.

Natural numbers can be used to count the cardinality of sets, as long as the set is finite.

Definition 5.2 (equal cardinality)**.** *X and Y have equal cardinality if and only if there exists a bijection $f : X \to Y$ from X to Y.*

Proposition 5.1 (properties of equal cardinality)**.**

- *X has equal cardinality with X.*
- *If X has equal cardinality with Y, then Y has equal cardinality with X.*

- *If X has equal cardinality with Y and Y has equal cardinality with Z, then X has equal cardinality with Z.*

Definition 5.3 (cardinality n)**.** *A set X has cardinality n or n elements if and only if it has equality cardinality with $\{i \in \mathbb{N} : 1 \leq i \leq n\}$.*

Proposition 5.2 (uniqueness of cardinality)**.**

Definition 5.4 (finite sets)**.**

Theorem 5.1 ($\mathbb{N}$ is infinite)**.** *The set of natural number $\mathbb{N}$ is an infinite set.*

Proof. □

Proposition 5.3 (cardinal arithmetic)**.**

6 Integers

Definition 6.1 ($\mathbb{Z}$)**.** *An integer is an expression of the form $a - b$, where a and b are natural numbers. Two integers are considered to be equal $a - b = c - d$ if and only if $a + d = c + b$.*

Definition 6.2 (sum)**.**

$$(a - b) + (c - d) := (a + c) - (b + d) \tag{6.1}$$

Definition 6.3 (product)**.**

$$(a - b) \times (c - d) := (ac + bd) - (ad + bc) \tag{6.2}$$

Lemma 6.1 (substitution axiom verified)**.** *Let a, b, a', b', c, d be natural numbers. If $(a - b) = (a' - b')$, then $(a - b) + (c - d) = (a' - b') + (c - d)$ and $(a - b) \times (c - d) = (a' - b') \times (c - d)$, and also $(c - d) + (a - b) = (c - d) + (a' - b')$ and $(c - d) \times (a - b) = (c - d) \times (a' - b')$.*

Definition 6.4 (negation)**.** *If $a - b$ is an integer, we define the negation $-(a - b)$ to be $(b - a)$. In particular if $n = n - 0$ is a positive natural number, we define its negation $-n = 0 - n$.*

Lemma 6.2 (trichotomy of integers)**.** *Let x be an integer. Then exactly one of the following three statement is true: (a) x is 0; (b) x is equal to a positive natural number n; (c) x is the negation $-n$ of a positive natural number n.*

Proposition 6.1 (commutative ring)**.** *Let x, y, z be integers, then*

$$x + y = y + x \tag{6.3}$$
$$(x + y) + z = x + (y + z) \tag{6.4}$$
$$x + 0 = 0 + x = x \tag{6.5}$$
$$x + (-x) = (-x) + x = 0 \tag{6.6}$$
$$xy = yx \tag{6.7}$$
$$(xy)z = x(yz) \tag{6.8}$$
$$x1 = 1x = x \tag{6.9}$$
$$x(y + z) = xy + xz \tag{6.10}$$
$$(y + x)x = yx + zx \tag{6.11}$$

Definition 6.5 (substraction)**.**

$$x - y := x + (-y) \tag{6.12}$$

Proposition 6.2 (zero divisor)**.** *Let a and b be integers such that $ab = 0$. Then either $a = 0$ or $b = 0$ (or both).*

Proposition 6.3 (cancellation law)**.** *If a, b, c are integers such that $ac = bc$ and c is non-zero, then $a = b$.*

Definition 6.6 (order)**.** *Let n and m be integers. $n \geq m$ or $m \leq n$ if and only if $n = m + a$ for some natural number a. $n > m$ or $n < m$ if and only if $n \geq m$ and $n \neq m$.*

Lemma 6.3 (properties of order). *Let a, b, c be integers, then*

- *$a > b$ if and only if $a - b$ is a positive natural number.*
- *(addition preserves order) If $a > b$, then $a + c > b + c$.*
- *(positive multiplication preserves order) If $a > b$ and c is positive, then $ac > bc$.*
- *(negation reverses order) If $a > b$, then $-a < -b$.*
- *(order is transitive) If $a > b$ and $b > c$, then $a > c$.*
- *(order trichotomy) Exactly one of the statements $a > b$, $a < b$, or $a = b$ is true.*

7 Rationals

Definition 7.1 ($\mathbb{Q}$). *A rational number is an expression of the form $a//b$, where a and b are integers and $b \neq 0$. $a//0$ is not considered to be a rational number. $a//b = c//d$ if and only if $ad = cb$.*

Definition 7.2 (sum).

$$(a//b) + (c//d) := (ad + bc)//(bd) \tag{7.1}$$

Definition 7.3 (multiplication).

$$(a//b) \times (c//d) := (ac)//(bd) \tag{7.2}$$

Definition 7.4 (negation).

$$-(a//b) := (-a)//b \tag{7.3}$$

Lemma 7.1 (substitution axiom verified). *Let a, b, c, d, a', b' be natural numbers. If $a//b = a'//b'$, then $a//b + c//d = a'//b' + c//d$, and $(a//b) \times (c//d) = (a'//b') \times (c//d)$, and $-(a//b) = -(a'//b')$.*

Definition 7.5 (reciprocal). *If $x = a//b$ is a non-zero rational (so that $a, b \neq 0$), then*

$$x^{-1} := b//a \tag{7.4}$$

Proposition 7.1 (field). *Let x, y, z be rationals.*

$$x + y = y + z \tag{7.5}$$
$$(x + y) + z = x + (y + z) \tag{7.6}$$
$$x + 0 = 0 + x = x \tag{7.7}$$
$$x + (-x) = (-x) + x = 0 \tag{7.8}$$
$$xy = yx \tag{7.9}$$
$$(xy)z = x(yz) \tag{7.10}$$
$$x1 = 1x = x \tag{7.11}$$
$$x(y + z) = xy + xz \tag{7.12}$$
$$(y + z)x = yx + zx \tag{7.13}$$
$$xx^{-1} = x^{-1}x = 1 \text{ for } x \neq 0 \tag{7.14}$$

Definition 7.6 (quotient). *Let x and y are two rationals and $y \neq 0$, then*

$$x/y := x \times y^{-1} \tag{7.15}$$

Definition 7.7 (substraction).

$$x - y := x + (-y) \tag{7.16}$$

Definition 7.8 (positive and negative). *A rational number x is positive if and only if $x = a//b$ for some positive integers a and b. It is said to be negative if and only if $x = -y$ for some positive rational y.*

Lemma 7.2 (trichotomy of rationals). *Let x be a rational number. Then exactly one of the following three statements is true: (a) $x = 0$; (b) x is a positive rational number; (c) x is a negative rational number.*

Definition 7.9 (order)**.** *Let x and y be rational numbers. $x > y$ if and only if $x - y$ is a positive rational number, and $x < y$ if and only if $x - y$ is a negative rational number. $x \geq y$ if and only if either $x > y$ or $x = y$.*

Proposition 7.2 (ordered field)**.** *Let x, y, z be rational numbers, then*

- *(order trichotomy) Exactly one of the three statements $x = y$, $x < y$, $x > y$ is true.*
- *(antisymmetric) $x < y$ if and only if $y > x$.*
- *(transitive) If $x < y$ and $y < z$, then $x < z$.*
- *(addition preserves order) If $x < y$, then $x + z < y + z$.*
- *(positive multiplication preserves order) If $x < y$ and z is positive, then $xz < yz$.*

7.1 Absolute Value

Definition 7.10 (absolute value)**.** *If x is a rational number, then the absolute value $|x|$ of x is defined as $|x| := x$ if x is positive; $|x| := -x$ if x is negative; $|x| := 0$ if $x = 0$.*

Definition 7.11 (distance)**.** *Let x and y be rational numbers.*

$$d(x, y) := |x - y| \tag{7.17}$$

Proposition 7.3 (properties of absolute value)**.** *Let x, y, z be rational numbers.*

- *(non-degeneracy) $|x| = 0$ if and only if $x = 0$.*
- *(triangle inequality) $|x + y| \leq |x| + |y|$.*
- *$-y \leq x \leq y$ if and only if $y \geq |x|$. In particular, $-|x| \leq x \leq |x|$.*
- *(multiplicativity) $|xy| = |x||y|$. In particular, $|-x| = |x|$.*

Proposition 7.4 (properties of distance)**.**

- *(non-degeneracy) $d(x, y) \geq 0$. $d(x, y) = 0$ if and only if $x = y$.*
- *(symmetry) $d(x, y) = d(y, x)$.*
- *(triangle inequality) $d(x, z) \leq d(x, y) + d(y, z)$.*

Proof.

- $d(x, z) = |x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z| = d(x, y) + d(y, z)$.
  $d(x, z) = d(x - y, z - y) = |(x - y) - (z - y)| \leq |x - y| + |z - y| = d(x, y) + d(z, y)$.

□

Definition 7.12 (ε-closeness)**.** *Let $\varepsilon > 0, x, y$ be rational numbers. y is ε-close to x if and only if $d(y, x) \leq \varepsilon$.*

Proposition 7.5 (properties of ε-close)**.** *Let x, y, z, w be rational numbers.*

- *If $x = y$, then x is ε-close to y for every $\varepsilon > 0$. Conversely, if x is ε-close to y for every $\varepsilon > 0$, then $x = y$.*
- *Let $\varepsilon > 0$. If x is ε-close to y, then y is ε-close to x.*
- *Let $\varepsilon, \delta > 0$. If x is ε-close to y, and y is δ-close to z, then x and z are $(\varepsilon + \delta)$-close.*

  Proof. $d(x, z) \leq d(x, y) + d(y, z) \leq \varepsilon + \delta$. □

- *Let $\varepsilon, \delta > 0$. If x and y are ε-close, and z and w are δ-close, then $x + z$ and $y + w$ are $(\varepsilon + \delta)$-close, and $x - z$ and $y - w$ are also $(\varepsilon + \delta)$-close.*

Proof. $d(x+z, y+w) \leq d(x+z, y+z) + d(y+z, y+w) = d(x,y) + d(z,w) \leq \varepsilon + \delta$. □

- *Let $\varepsilon > 0$. If x and y are ε-close, then they are also ε'-close for every $\varepsilon' > \varepsilon$.*

  Proof. $d(x,y) \leq \varepsilon < \varepsilon'$. □

- *Let $\varepsilon > 0$. If y and z are both ε-close to x, and w is between y and z (i.e. $y \leq w \leq z$ or $z \leq w \leq y$), then w is also ε-close to x.*

  Lemma 7.3. *Let x, y, z be rational numbers, such that $x \leq y \leq z$. Then either $|y| \leq |x|$ or $|y| \leq |z|$ (or both).*

  Proof.

  - If $y = 0$, then $|y| = 0 \leq |x|$ and $|y| = 0 \leq |z|$.
  - If $y > 0$, then $0 < y \leq z$, then $|y| \leq |z|$.
  - If $y < 0$, then $-y \leq -x < 0$, then $|y| \leq |x|$.

  □

  Proof. Suppose $y \leq w \leq z$, then

  $$y - x \leq w - x \leq z - x$$

  Then either

  $$|w - x| \leq |y - x|$$

  or

  $$|w - x| \leq |z - x|$$

  Then either

  $$d(w, x) \leq d(y, x) \leq \varepsilon$$

  or

  $$d(w, x) \leq d(z, x) \leq \varepsilon$$

  □

- *Let $\varepsilon > 0$. If x and y are ε-close, and z is non-zero, then xz and yz are $\varepsilon|z|$-close.*

  Proof. $d(xz, yz) = |xz - yz| = |z||x - y| \leq |z|\varepsilon$. □

- *Let $\varepsilon, \delta > 0$. If x and y are ε-close, and z and w are δ-close, then xz and yw are $(\varepsilon|z| + \delta|x| + \varepsilon\delta)$-close.*

  Proof.

  $$d(xz, yw) \leq d(xz, yz) + d(yz, yw) \leq |z|d(x, y) + |y|d(z, w)$$

  Since

  $$d(y, 0) \leq d(y, x) + d(x, 0)$$

  then

  $$d(xz, yw) \leq |z|d(x, y) + (d(y, x) + |x|)d(z, w) \leq |z|\varepsilon + \varepsilon\delta + |x|\delta$$

  □

7.2 Exponentiation

Definition 7.13 (exponentiation)**.** *Let x be a rational number. $x^0 := 1$. In particular, $0^0 := 1$. $x^{n+1} := x^n \times x$.*

Proposition 7.6 (properties of exponentiation)**.** *Let x and y be rational numbers.*

- *$x^n x^m = x^{n+m}$, $(x^n)^m = x^{nm}$, and $(xy)^n = x^n y^n$.*
- *Suppose $n > 0$. Then $x^n = 0$ if and only if $x = 0$.*
- *If $x \geq y \geq 0$, then $x^n \geq y^n \geq 0$. If $x > y \geq 0$ and $n > 0$, then $x^n > y^n \geq 0$.*
- *$|x^n| = |x|^n$.*

Definition 7.14 (negative exponentiation)**.** *Let x be a non-zero rational number. Then for any negative integer $-n$, $x^{-n} := 1/x^n$.*

Proposition 7.7 (properties of exponentiation)**.** *Let x and y be non-zero rational numbers, and let n and m be integers.*

- *$x^n x^m = x^{n+m}$, $(x^n)^m = x^{nm}$, and $(xy)^n = x^n y^n$.*
- *If $x \geq y > 0$, then $x^n \geq y^n > 0$ if n is positive, and $0 < x^n < y^n$ if n is negative.*
- *If $x, y > 0, n \neq 0, x^n = y^n$, then $x = y$*
- *$|x^n| = |x|^n$.*

7.3 Gaps in Rational Numbers

Proposition 7.8 (integer part)**.** *Let x be a rational number. Then there exists a unique integer n such that $n \leq x < n + 1$. In particular there exists a natural number N such that $N > x$.*

Proof. Existence:

- If $x = 0$, then $0 \leq x < 1$.
- If $x > 0$, then suppose $x = a/b = (mb + r)/b = m + r/b$, where $0 \leq r < b$. Then $m \leq x < m + 1$.
- If $x < 0$, then suppose $x = -m - r/b$. Then $-m - 1 \leq x < -m$.

Uniqueness:
Suppose

$$n_1 \leq x < n_1 + 1$$

and

$$n_2 \leq x < n_2 + 1$$

Suppose $n_1 < n_2$, then

$$n_1 + 1 \leq n_2 \leq x < n_1 + 1$$

Contradict. □

Proposition 7.9 (intersperse)**.** *If x and y are two rational numbers such that $x < y$. Then there exists a third rational z such that $x < z < y$.*

Proof. $z = (x + y)/2$. □

Definition 7.15 (infinite descent)**.** *A sequence $a_0, a_1, a_2, \ldots$ of numbers is said to be infinite descent if $a_n > a_{n+1}$ for all natural numbers n.*

Lemma 7.4 (principle of infinite descent)**.** *It is not possible to have a sequence of natural numbers which is infinite descent.*

Proof. Suppose for sake of contradiction that $a_0 > a_1 > a_2 > \dots$ is a sequence of natural numbers which is infinite descent, then $a_n \geq k$ for all $k \in \mathbb{N}$ and all $n \in \mathbb{N}$.

When $k = 0$, $a_n \geq 0$ for all $n \in \mathbb{N}$.

Suppose $a_n \geq k$ for all $n \in \mathbb{N}$.

Then

$$a_n > a_{n+1} \geq k$$
$$a_n \geq k + 1$$

This closes the induction.

When $k = a_n + 1$,

$$a_n \geq a_n + 1$$

Contradiction. □

Proposition 7.10 ($\sqrt{2}$)**.** *There does not exist any rational number x for which $x^2 = 2$.*

Proof. Suppose $x = p/q$, where p, q are natural numbers.

Then

$$x^2 = p^2/q^2 = 2$$
$$p^2 = 2q^2$$

$p > q$ and p is even.

Let $p = 2k$, then

$$4k^2 = 2q^2$$
$$q^2 = 2k^2$$

$q > k$ and q is even.

Repeat this process

$$p > q > k > ...$$

Contradict with principle of infinite descent. □

Proposition 7.11. *For every rational number $\varepsilon > 0$, there exists a non-negative rational number x such that $x^2 < 2 < (x + \varepsilon)^2$.*

Proof. When $\varepsilon \geq 1$,

$$1^2 < 2 < (1 + \varepsilon)^2.$$

When $0 < \varepsilon < 1$, suppose for sake of contradiction that there is no x for which $x^2 < 2 < (x + \varepsilon)^2$.

Then for every $x^2 < 2$,

$$(x + \varepsilon)^2 < 2$$

Let $x = 0$, then

$$(0 + \varepsilon)^2 = \varepsilon^2 < 2$$

Suppose $(n\varepsilon)^2 < 2$, then

$$(n\varepsilon + \varepsilon)^2 = ((n + 1)\varepsilon) < 2$$

This closes induction.

Let $n > 2/\varepsilon$, then

$$(n\varepsilon)^2 > 4$$

Contradiction. □

8 Real Numbers

8.1 Cauchy Sequences

Definition 8.1 (sequence)**.** *A sequence $(a_n)_{n=m}^{\infty}$ of rational numbers is a collection of rationals $a_m, a_{m+1}, a_{m+2}, \ldots$.*

Definition 8.2 (ε-steadiness)**.** *Let $\varepsilon > 0$. The sequence $a_0, a_1, a_2, \ldots$ is ε-steady if and only if $d(a_j, a_k) \leq \varepsilon$ for all j, k.*

Definition 8.3 (eventual ε-steadiness)**.** *Let $\varepsilon > 0$. The sequence $a_0, a_1, a_2, \ldots$ is eventual ε-steady if and only if there exists a natural number $N \geq 0$ such that $d(a_j, a_k) \leq \varepsilon$ for all $j, k \geq N$.*

Definition 8.4 (Cauchy sequence)**.** *A sequence $(a_n)_{n=0}^{\infty}$ is a Cauchy sequence if and only if for every rational $\varepsilon > 0$, the sequence $(a_n)_{n=0}^{\infty}$ is eventually ε-steady.*

Definition 8.5. *The sequence $(1/n)_{n=1}^{\infty}$ is a Cauchy sequence.*

Proof. For all $j, k \geq N$,

$$0 < 1/j, 1/k \leq 1/N$$
$$|1/j - 1/k| \leq 1/N$$

Choose an N such that $1/N \leq \varepsilon$ or $N \geq 1/\varepsilon$. □

Definition 8.6 (bounded sequence)**.** *Let $M \geq 0$ be rational. A finite sequence $a_1, a_2, \ldots, a_n$ is bounded by M if and only if $|a_i| \leq M$ for all $1 \leq i \leq n$. A infinite sequence $(a_n)_{n=1}^{\infty}$ is bounded by M if and only if $|a_i| \leq M$ for all $i \geq 1$. A sequence is bounded if and only if it is bounded by M for some rational $M \geq 0$.*

Lemma 8.1 (finite sequences are bounded)**.** *Every finite sequence $a_1, a_2, \ldots, a_n$ is bounded.*

Proof. When $n = 1$,

$$|a_1| \leq |a_1|$$

Suppose $a_1, a_2, \ldots, a_n$ is bounded by M, then

$$|a_i| \leq M \text{ for all } 1 \leq i \leq n$$

Then

$$|a_i| \leq M + |a_{n+1}| \text{ for all } 1 \leq i \leq n+1$$

This closes induction. □

Lemma 8.2 (Cauchy sequences are bounded)**.** *Every Cauchy sequence $(a_n)_{n=1}^{\infty}$ is bounded.*

Proof. For every $\varepsilon > 0$, there exists a natural number $N \geq 1$ such that $|a_j - a_k| \leq \varepsilon$ for all $j, k \geq N$.
Then

$$d(a_j, 0) \leq d(a_j, a_k) + d(a_k, 0) = |a_j - a_k| + |a_k| \leq \varepsilon + |a_k| \text{ for all } j \geq N$$

For $n < N$, sequence $a_1, a_2, \ldots, a_n$ is bounded. □

8.2 Equivalent Sequences

Definition 8.7 (ε-close sequences)**.** *Let $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ be two sequences, and let $\varepsilon > 0$. The sequence $a_0, a_1, a_2, \ldots$ is ε-close to the sequence $b_0, b_1, b_2, \ldots$ if and only if $d(a_n, b_n) \leq \varepsilon$ for all $n = 0, 1, 2, \ldots$.*

Definition 8.8 (eventually ε-close sequences)**.** *Let $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ be two sequences, and let $\varepsilon > 0$. The sequence $a_0, a_1, a_2, \ldots$ is eventually ε-close to the sequence $b_0, b_1, b_2, \ldots$ if and only if there exists an $N \geq 0$ such that $d(a_n, b_n) \leq \varepsilon$ for all $n \geq N$.*

Definition 8.9 (equivalent sequences)**.** *Two sequences $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ are equivalent if and only if for every rational $\varepsilon > 0$, the sequences $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ are eventually ε-close.*

Lemma 8.3.

$$10^{-m} < 10^{-n} \text{ whenever } m > n$$

Proof. Since $m > n$, then

$$m = n + k$$

When $k = 1$,

$$10^{-(n+1)} < 10^{-n}$$

Suppose

$$10^{-(n+k)} < 10^{-n} \tag{8.1}$$

Then

$$10^{-(n+k+1)} < 10^{-n}$$

This closes induction. □

Lemma 8.4.

$$10^{-n} \leq 1/n \text{ for } n > 0$$

Proof. When $n = 1$,

$$10^{-1} \leq 1$$

Suppose

$$10^{-n} \leq 1/n$$

Then

$$10^{-(n+1)} \leq 1/n < 1/(n+1)$$

This closes induction. □

Proposition 8.1. *Let $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ be the sequences $a_n = 1 + 10^{-n}$ and $b_n = 1 - 10^{-n}$. Then the sequences a_n, b_n are equivalent.*

Proof. For all $n \geq N$,

$$d(a_n, b_n) = |a_n - b_n| = 2 \times 10^{-n} \leq 2 \times 10^{-N}$$

Choose N such that

$$2 \times 10^{-N} \leq 2/N \leq \varepsilon$$

□

8.3 Real Numbers

Definition 8.10 (irrational number)**.**

Definition 8.11 (transcendental number)**.** π, e

9 Limits of Sequences

10 Power Series

11 Infinite Sets

12 Continuous Functions

13 Differentiation of Functions

14 Riemann Integral

A Tools

Definition A.1 (implication)**.** *"If P, then Q" = "If not Q, then not P".*

Lemma A.1 (de mongan laws)**.**

- *not "A and B" = "not A or not B"*
- *not "A or B" = "not A and not B"*

Axiom A.1 (subsititution axiom)**.** *Given any two objects x and y, if $x = y$, then $f(x) = f(y)$ for all functions or operations. For any property $P(x)$ pertaining to x, if $x = y$, then $P(x)$ and $P(y)$ are equivalent statements.*

Definition A.2 (conjunction and disjunction)**.**

- *The statement "X and Y" is true if X and Y are both true.*
- *The statement "X or Y" is true if either X or Y is true, or both.*

References

[1] Terence Tao. *Analysis I.* Vol. 1. Springer, 2022. ISBN: 978-981-19-7261-4.

[2] Terence Tao. *Analysis II.* Vol. 2. Springer, 2022. ISBN: 978-981-19-7284-3.